

MODULE 23 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Observability Foundations

Domain 3 begins. Improving what exists, starting with being able to see it.

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Monitoring tells you **that** something broke.
Observability lets you ask **why** without
shipping new code.

Metrics: resolution costs money

- ◆ **Default metrics are five-minute resolution and free.** EC2 detailed monitoring gives one minute, and is charged.
- ◆ **High-resolution custom metrics go to one second**, which is what makes sub-minute alarms possible.
- ◆ Publish custom metrics with PutMetricData, or with **embedded metric format** in a log line, which avoids a separate API call.
- ◆ **Memory and disk utilisation are not default EC2 metrics.** They need the CloudWatch agent installed.

ON THE EXAM

An alarm cannot be faster than the metric feeding it. A one-minute alarm on a five-minute metric is a five-minute alarm.

Logs, and where the money goes

- ◆ **Retention defaults to never expire.** This is the single most common CloudWatch cost leak in any estate.
- ◆ **The Infrequent Access log class** costs less for logs that are stored for compliance and rarely queried.
- ◆ **Metric filters** turn log text into a numeric metric, so you can alarm on something an application only prints.
- ◆ **Subscription filters** stream logs to Kinesis, Data Firehose or Lambda for real-time processing or archiving.

ON THE EXAM

Metric filters are how you alarm on log content. Without one, a log line saying payment failed cannot trigger anything.

Querying: three tools, one question

- ◆ **Logs Insights** for recent operational logs, queried occasionally. No setup, pay per query.
- ◆ **Athena over S3** for years of archived logs, queried rarely and cheaply.
- ◆ **OpenSearch** when people search constantly and need dashboards. You pay to index everything, always.
- ◆ The deciding question is **query frequency**, not data volume. Constant searching justifies an index; occasional searching does not.

ON THE EXAM

Same as module ten: occasional queries mean Athena or Logs Insights. Constant interactive search means OpenSearch.

The alarm types

TYPE	FIRES ON	USE IT WHEN
Metric alarm	A metric crossing a static threshold	You know what normal looks like
Anomaly detection	Deviation from a learned band	Normal varies by hour or day
Composite alarm	A boolean of other alarms	Reducing noise, or requiring agreement
Metric math alarm	An expression across metrics	Error rate, not error count

ON THE EXAM

Metric math is the underused one. Alarming on errors divided by requests survives traffic growth; alarming on error count does not.

Two settings that decide whether an alarm works

- ◆ **Missing data treatment.** The default is to treat missing data as missing, so an alarm may simply never evaluate.
- ◆ If a metric stops being published when the thing breaks, **treat missing data as breaching** or the alarm stays silent.
- ◆ **Datapoints to alarm** versus evaluation periods lets you require, say, three bad minutes out of five. That is how you stop flapping.
- ◆ **Composite alarms** also support suppression, so a known dependency failure does not page five teams.

ON THE EXAM

An alarm that has never fired is not evidence of health. A metric that disappears on failure is the classic silent alarm.

Seeing across accounts

- ◆ **Cross-account observability** nominates a monitoring account that sees metrics, logs and traces from source accounts.
- ◆ From there you can run **cross-account Logs Insights queries** and create **alarms on other accounts' metrics**.
- ◆ Anomaly detection also works on cross-account metrics, so one team can watch behaviour across a whole estate.
- ◆ Pair it with the delegated administrator pattern from module ten. The monitoring account is not the management account.

ON THE EXAM

Without this, an incident spanning four accounts means four consoles and no single timeline. That is the problem it removes.

From metrics to service level objectives

- ◆ **An SLI is a measurement.** An SLO is the target for it. An SLA is the contractual promise, usually looser than the SLO.
- ◆ **Application Signals** collects SLIs automatically from instrumented applications and tracks SLOs, with burn-rate alerts.
- ◆ It also builds a service map with drill-down to metrics, logs and traces, and works across accounts.
- ◆ Newer than most study material. Where a question is generic, **metric math on error rate and latency** is also a defensible answer.

ON THE EXAM

Set the internal SLO tighter than the external SLA. That gap is your warning time before a contractual breach.

Reading the scenario

THE SCENARIO SAYS

Memory or disk metrics are missing

One failure triggers dozens of alarms

The alarm never fired during the outage

Alarm on a message the app writes to a log

One view of an incident across many accounts

CloudWatch Logs cost is growing steadily

IT IS POINTING AT

The CloudWatch agent

A composite alarm

Missing data treatment

A metric filter

Cross-account observability

Retention, and the IA log class

ON THE EXAM

Ask what the team could not see, and why. Most Domain 3 questions start with a visibility gap rather than a broken component.

Meridian Freight's fleet of EC2 instances runs out of memory roughly once a fortnight, causing a crash. The team wants to scale out before that happens, but CloudWatch shows no memory metric to alarm on. CPU stays below 30% throughout.

The instances run a standard Amazon Linux AML.

What should the solutions architect recommend?

- ☐ A Enable EC2 detailed monitoring to expose memory metrics at one-minute resolution
- ☐ B Install the CloudWatch agent to publish memory as a custom metric, and alarm on it
- ☐ C Alarm on CPU utilisation as a proxy for memory pressure
- ☐ D Enable Container Insights on the instances

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Detailed monitoring

The sharpest distractor. Detailed monitoring changes resolution from five minutes to one, and adds no new metrics at all.

C CPU as a proxy

The scenario says CPU stays below 30% while memory exhausts. It is explicitly not correlated here.

B Install the CloudWatch agent

Memory lives inside the guest operating system, which the hypervisor cannot see. The agent is the only way to publish it.

D Container Insights

For ECS and EKS container metrics. Nothing suggests containers are involved.

THE SIGNAL

Detailed monitoring buys frequency, not coverage. Anything inside the guest OS needs an agent, and always has.

When the primary database becomes slow, 47 alarms fire within two minutes: application latency, queue depth, error rates, load balancer 5xx counts and Lambda durations. The on-call engineer receives 47 pages and spends the first ten minutes working out which one matters.

All 47 alarms are individually correct.

What should the solutions architect implement?

- ☐ A Raise the thresholds on the 46 downstream alarms so they fire less often
- ☐ B Composite alarms that group the downstream signals, with suppression when the database alarm is active
- ☐ C Delete the downstream alarms and keep only the database alarm
- ☐ D Route all alarms to a single SNS topic

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Raise the thresholds

Makes 47 correct alarms less sensitive, so they also miss real problems. The alarms are not wrong; the paging is.

C Delete the downstream alarms

Those alarms are the only signal when the database is fine and something downstream is not. It trades noise for blindness.

B Composite alarms with suppression

Groups the symptoms, and suppression means a known upstream cause pages once instead of 47 times.

D One SNS topic

Delivers 47 messages to one place. The engineer still has 47 things to read and rank.

THE SIGNAL

The alarms are all correct, which is stated deliberately. That eliminates every option that makes them less sensitive.

A critical batch job publishes a custom metric on each successful run. An alarm is configured to fire if that metric falls below one in an hour. Last week the job's host failed and no runs happened for eleven hours. The alarm never fired.

The alarm configuration was reviewed and the threshold is correct.

What is the cause?

- ☐ A The metric stopped being published, and missing data is treated as missing by default
- ☐ B The alarm needed a longer evaluation period
- ☐ C Custom metrics cannot be used in alarms
- ☐ D The alarm needed detailed monitoring enabled

◆ PAUSE AND DECIDE

What is the cause?

- A

 Missing data treated as missing
With no runs there is no data point, so the alarm has nothing to evaluate and stays in insufficient data. Set missing data to breaching.
- B

 Longer evaluation period
Evaluating over longer still evaluates nothing. The absence of data points is the problem, not their number.
- C

 Custom metrics cannot alarm
They can, and routinely do. The alarm existed and was correctly configured.
- D

 Detailed monitoring
Applies to EC2 metric resolution. It has no bearing on a custom metric published by an application.

THE SIGNAL

The metric that proves a thing is working disappears when it stops working. That is the trap, and it needs missing-data set to breaching.

An incident spans four accounts: a front-end account, an API account, a shared services account and a data account. Engineers currently switch roles between all four, comparing timestamps by hand, and typically take 40 minutes to build a coherent timeline.

There are 200 accounts in the organisation in total.

What should the solutions architect implement?

- ☐ A CloudWatch cross-account observability, with a dedicated monitoring account
- ☐ B Copy all logs to a central S3 bucket and query them with Athena
- ☐ C Give every engineer a role in all 200 accounts
- ☐ D Run all workloads in a single account to simplify monitoring

◆ PAUSE AND DECIDE

What should the solutions architect implement?

- A

Cross-account observability
One monitoring account sees metrics, logs and traces from all sources, with cross-account Logs Insights queries and alarms.
- B

Central S3 plus Athena
Good for archived forensics and poor for a live incident. It also covers logs only, leaving metrics and traces behind.
- C

Roles everywhere
Faster switching between 200 consoles. The problem is the switching itself, not the permission to switch.
- D

One account
Discards the isolation, quota separation and blast-radius boundaries the organisation deliberately built.

THE SIGNAL

The complaint is a fragmented timeline, not missing data. Every account already has the data; nothing correlates it.

CloudWatch Logs charges have grown to \$38,000 a month. Investigation finds 4,200 log groups, of which 3,900 have no retention setting. Compliance requires seven years of application logs. Engineers query logs from the last fortnight almost daily, and older logs perhaps twice a year.

No log group has ever been deleted.

What should the solutions architect recommend?

- ☐ A Set retention on all log groups to 14 days
- ☐ B Set short retention, and export or subscribe older logs to S3 with lifecycle policies for the seven-year requirement
- ☐ C Move every log group to the Infrequent Access class
- ☐ D Reduce application log verbosity

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 14 days everywhere
Cuts the bill and destroys the seven-year compliance requirement. It solves cost by creating a much worse problem.
- B

 Short retention plus S3 archive
Recent logs stay queryable in CloudWatch, older logs move to far cheaper S3 storage, and Athena serves the twice-a-year question.
- C

 IA class everywhere
Helps, and IA is still CloudWatch Logs pricing. It is a discount, not the order-of-magnitude change S3 with lifecycle gives.
- D

 Less verbose logging
Reduces future volume and does nothing about years of accumulated data with no retention set.

THE SIGNAL

Two requirements: cheap seven-year retention, and fast access to a fortnight. One store cannot serve both well.

A company promises customers 99.9% availability contractually. Today it alarms when 5xx error count exceeds 100 in five minutes. As traffic has tripled, the alarm now fires during normal operation, and during a quiet-period outage it did not fire at all.

The team needs alarming that reflects the promise as traffic changes.

What should the solutions architect recommend?

- ☐ A Raise the threshold from 100 to 300 to match the tripled traffic
- ☐ B Alarm on an error rate using metric math, and track it as an SLO against a tighter internal target
- ☐ C Use an anomaly detection alarm on 5xx count
- ☐ D Alarm on CPU utilisation of the web tier instead

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Raise to 300
Correct today and wrong again next year. A count threshold has to be re-tuned every time traffic changes.
- B** Error rate via metric math, tracked as an SLO
A ratio is traffic-independent, so it works at any volume, and it maps directly onto the 99.9% promise.
- C** Anomaly detection on count
Learns normal and still measures the wrong thing. A small absolute error count during a quiet period can still breach 99.9%.
- D** CPU utilisation
A resource metric standing in for a customer-facing one. An outage returning errors quickly may use very little CPU.

THE SIGNAL

Both failures come from alarming on a count. The quiet-period miss and the busy-period noise are the same bug.

What to carry out of this module

- ◆ **Detailed monitoring buys frequency, not coverage.** Memory and disk need the CloudWatch agent.
- ◆ **Missing data defaults to missing**, so an alarm on a metric that disappears on failure stays silent.
- ◆ **Alarm on rates, not counts.** A ratio survives traffic growth; a count threshold needs re-tuning forever.
- ◆ Composite alarms with suppression turn many symptoms of one cause into one page.

ON THE EXAM

Query frequency, not data volume, decides between Logs Insights, Athena and OpenSearch. Constant searching justifies an index.